TAPat: Hackathon Build Plan

Nov 4, 2026 · @jim

Pitch and problem

TAPat lets an AI agent release, cap, and claw back fuel aid for tricycle drivers, enforced on-chain by Noir and settled through Velo. Verify once, pay many times. No queues, no ghosts, no expired funds.

The 2026 fuel subsidy showed the trade-off every aid program faces: safe and slow, or fast and leaky.

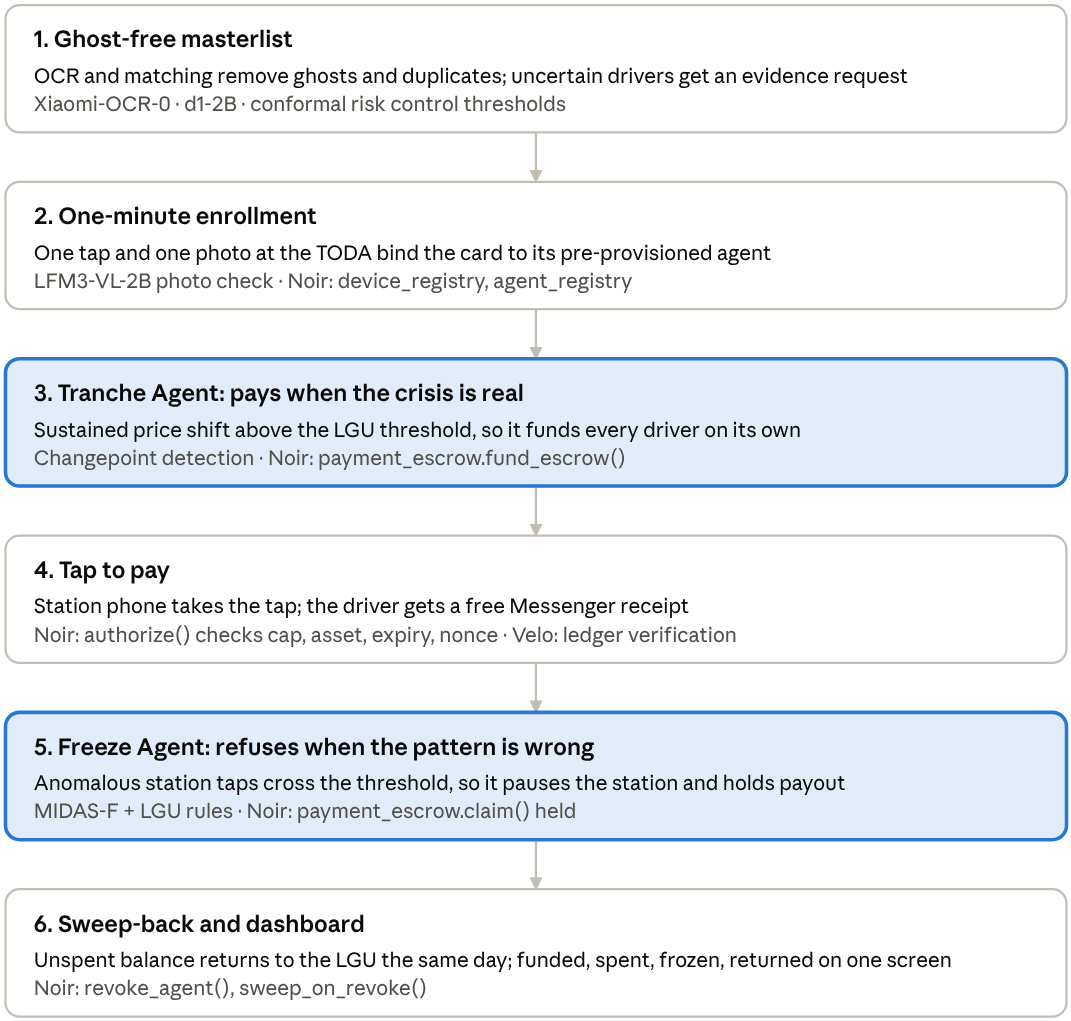
- **Scale:** 723,000 tricycle drivers slated for ₱5,000 each (PNA).
- **Leakage:** ₱2.7B nearly lost to duplicate and ghost beneficiaries (DSWD via Daily Tribune).
- **Unpaid and expired:** ₱709.88M in DOTr subsidies never reached drivers, and leftovers returned to the treasury after two years (COA via Astig.ph, NET25).
- **Human cost:** DSWD kept payouts in person to stop ghosts, and drivers fainted in heat-exposed queues.

The root cause: money moves as one lump sum, with no rules attached and no way to recover it until an audit a year later.

How it works

Six features, one loop: a clean list in, a driver enrolled in a minute, and an agent that funds, refuses, and returns money.

Two agents make the money decisions; Noir enforces them; Velo proves they settled



TAPat loop · 6 features

1. **Ghost-free masterlist.** Xiaomi-OCR-0 reads the LGU list on an office laptop. d1-2B scores same-person matches across name variants. Conformal risk control sets the approve and reject thresholds so wrongful denials stay under a rate the LGU picks. Fixed rules always block: duplicate plate, age under 18, one plate on several names. Uncertain drivers get an automatic Messenger request for an OR/CR photo, so there is no caseworker queue.
2. **One-minute enrollment.** At the TODA, one tap binds the card and one photo lets LFM3-VL-2B match the plate and TODA sticker to the record. The card's Noir agent is already provisioned with a per-tap cap, a PHP-pegged asset, and the program expiry.

3. **Tranche Agent: pays when the crisis is real.** Watches the DOE pump price. When changepoint detection confirms a sustained shift and the price is above the LGU threshold, it calls `fund_escrow()` for every driver inside the budget the LGU signed. Every decision is logged with its numbers.
4. **Tap to pay.** Any NFC Android phone is the station terminal. The attendant enters the amount, the driver taps, Noir `authorize()` enforces cap, asset, expiry, and nonce, and Velo marks the payment paid only after ledger verification. Velo Gas Station pays all fees, and the driver gets a free Messenger receipt.
5. **Freeze Agent: refuses when the pattern is wrong.** Every tap is a card-to-station edge scored by MIDAS-F in real time, alongside LGU rule thresholds: charges near the cap every time, or volume far above the station's normal. Crossing a threshold pauses the station and holds its `claim()`.
6. **Sweep-back and dashboard.** At program end, `revoke_agent()` and `sweep_on_revoke()` return unspent funds to the LGU the same day. One dashboard shows funded, spent by station, frozen, and returned, exportable for COA.

Not in TAPat: PDAX cash-out (UAT only) and any privacy-credential claim.

Hackathon fit

Two agents make the payment decisions, and all three Noir contracts plus Velo are used for what they were built for.

Requirement	How TAPat meets it
Agentic payment	Tranche Agent funds (`fund_escrow()`); Freeze Agent withholds settlement (holds `claim()`); card agents request payment (`authorize()`)
Built with Velo	Checkout session per tap, ledger verification, signed webhooks, Gas Station
Noir `device_registry`	One registered device per driver card, owned by the LGU
Noir `agent_registry`	Per-card cap, asset, and expiry; `check_payment()`
Noir `payment_escrow`	`fund_escrow()` , `authorize()` , `claim()` , `sweep_on_revoke()`

Pitch angle: Noir's own docs say the escrow `authorize()` and `claim()` flow isn't wired into its app. TAPat wires it end to end.

Algorithms from arXiv

Three published algorithms give every automatic decision a number COA can audit.

Algorithm	Decision it makes
Conformal Risk Control	Approve or reject a beneficiary, with a bounded wrongful-denial rate
Bayesian Online Changepoint Detection	Fund a tranche only on a sustained price shift
MIDAS-F	Freeze a station whose tap pattern is anomalous

For the demo, calibrate the conformal thresholds on a synthetic labeled set, and feed changepoint detection and MIDAS-F simulated price and tap streams.

Demo script (about 3 minutes)

Lead with the agent's two money decisions, then show why the money reaches real people. Every number is on screen.

1. **Problem (20 s):** ₱2.7B nearly lost to ghosts, ₱709.88M never delivered, drivers fainting in queues.
2. **Agent funds (40 s):** A simulated DOE price series jumps. The changepoint probability climbs; once the shift is sustained and above threshold, the Tranche Agent funds all three escrows. Show the decision log and tx hashes.
3. **Tap and pay (30 s):** Driver 1 taps at the station phone; Velo verifies and the Messenger receipt arrives. Then an over-cap amount and an expired card are both rejected on-chain.
4. **Agent freezes (35 s):** A burst of simulated taps hits one station. The MIDAS-F score crosses the threshold, and the Freeze Agent pauses it and holds its `claim()`. An honest station claims normally.
5. **No ghosts, no queues (30 s):** Drop a messy scanned list. A ghost is caught, two uncertain drivers get an automatic evidence request, and one tap plus one photo enrolls a driver.

6. **Money comes back (20 s):** End the program. Sweep returns the unspent balance; dashboard shows funded, spent, frozen, returned.
7. **Close (10 s):** "The agent pays when the crisis is real, refuses when the pattern is wrong, and returns what's left."

Judge Q&A

Question	Answer
Where's the agent, really?	Two agents move money on their own: the Tranche Agent funds escrows, and the Freeze Agent withholds a station's settlement. Both act inside limits the LGU signed.
Is AI deciding who gets aid?	Inside an error bound the LGU sets. Conformal risk control caps wrongful denials, uncertain drivers are asked for evidence automatically, and every rejection can be appealed in one tap.
Do stations need new hardware?	No. Any NFC-capable Android phone runs the station app.
What stops a station from overcharging?	The per-tap cap, a Messenger receipt to the driver, and the Freeze Agent pausing stations with odd patterns.
NFC cards can be cloned.	One agent per card caps the damage to one driver's allocation.
Can you restrict spending to fuel?	Noir has no merchant allow-list, so only registered station accounts can create Velo checkout sessions.
What about data privacy?	OCR and matching run on the LGU laptop. Stations see a card hash and a balance, never names.
Is the fuel crisis over?	Fuel spikes recur, and the rail works for any AICS-style program.
Is this production-ready?	No. Testnet alpha, like Noir and Velo. Mainnet needs a security audit first.

Day-1 checklist (Thursday morning)

Clear these before writing app code; the seam check decides the architecture.

- [ ] **Registration tonight** (closes Nov 4, 11:59 PM): Google Form, both GitHub stars, Velo, Noir, @highrable, @GigPayPH, @VertigralTech followed, screenshots per member; project registered on velo-build.dev.
- [ ] **Velo and Noir seam:** can Velo verify a payment made through `authorize()` / `claim()`? If not, the card agent wallet pays the Velo intent directly and escrow handles funding, freeze, and sweep.
- [ ] **Latest Noir Testnet contract IDs**, and confirm `sweep_on_revoke()` is deployed.
- [ ] **Asset** that both the Noir policy and Velo checkout accept.
- [ ] **Gas Station** activated with the contracts on its allow-list.
- [ ] **Filipino names:** "Dela Cruz / De la Cruz / Delacruz" matched correctly by d1-2B.